

Effects of freebase/protonated nicotine concentration, liquid composition and electrical power on throat hit in direct-to-lung vaping: theory and clinical measurements

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ABSTRACT

Background For decades, the tobacco industry has engineered the sensory characteristics of combustible tobacco products including the degree of harshness experienced at the back of the throat. Commonly referred to as ‘throat hit’, this harshness derives from absorption of gas phase nicotine and other constituents by the sensory nerves. People who use tobacco products associate throat hit with the positive psychological effects of nicotine, making it a secondary reinforcer for smoking. On the other hand, high throat hit can make products aversive to naïve nicotine users and serves as a barrier to consumption of tobacco products.

Methods We developed a first-principles theoretical model to predict nicotine absorption in the throat as a function of electronic nicotine delivery systems (ENDS) device power, liquid composition and puffing topography. The predicted nicotine absorption was compared with subjective throat harshness reported by human participants. We also simulated several ENDS use scenarios to identify the most important processes and factors that govern nicotine absorption in the throat.

Results Across various ENDS configurations, we found that computed nicotine gas absorption in the throat was associated with subjective harshness scores ($r=0.62$; $p<0.00001$). We also found that liquid nicotine concentration, nicotine form, aerosol temperature and inhalation rate strongly influence nicotine absorption in the throat per unit of nicotine emitted by an ENDS.

Conclusions Nicotine throat hit can be predicted and manipulated through ENDS device and liquid design variables. Regulating ENDS throat hit is feasible and may help reduce product appeal to nicotine-naïve individuals while maintaining acceptability for smokers.

INTRODUCTION

Marketed as a less harmful alternative for cigarette smokers, electronic nicotine delivery systems (ENDS) appear to have arrested decades of declining nicotine product use among youth.^{1–4} Although combustible cigarette use remains on the decline, ENDS continue to be the most commonly used tobacco product among youth, despite recent data indicating a drop in current-use rates by US high-school students from 14.1% in 2022 to 10.0% in 2023.⁵

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Throat hit, the harsh sensation experienced at the back of the throat during smoking, is driven by gas phase nicotine absorption and reinforces drug administration. The tobacco industry in recent years began marketing electronic nicotine delivery systems (ENDS) products with protonated nicotine that lowers throat hit/harshness.

WHAT THIS STUDY ADDS

⇒ This study shows that computed nicotine gas absorption in the throat is associated with sensory harshness by ENDS users across a wide range of conditions and elucidates how common ENDS variables can be manipulated to influence nicotine emissions and nicotine gas uptake in the throat independently.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study points to the technical feasibility of regulating ENDS products both for nicotine emissions and sensory harshness, for example, to create a sensory barrier of entry for nicotine-naïve individuals while providing experienced smokers a product that sufficiently suppresses smoking withdrawal symptoms.

While ENDS were introduced in the early 2000s, the rapid rise in use among youth came much later with the marketing of products that contained high-strength, protonated (or ‘salt’) nicotine liquids.¹ These protonated nicotine products elicited far less throat irritation than comparable concentration freebase nicotine products⁶ and facilitated nicotine inhalation by previously nicotine-naïve individuals.⁷

Nicotine gas absorption in the region of the sensory nerves at the back of the throat causes a harsh sensation known as ‘throat hit’.⁸ Internal tobacco industry documents^{9–11} indicate that for decades, the industry has been aware of the role of gas phase nicotine in generating throat hit and studied the matter empirically. For example, a 1991 internal report⁹ by R.J. Reynolds cited various company studies of the processes of nicotine deposition in the throat and related subjective effects

in human participants. Among the studies reported was one in which smokers found that cigarettes with acids added to the tobacco during manufacturing elicited less throat hit than even low-nicotine content cigarettes without added acids. Adding acid to tobacco converts nicotine from its free-base form to its non-volatile protonated form and reduces the concentration of gas phase nicotine in the aerosol.^{8 12}

Interestingly, experienced smokers associate throat hit with the positive psychological effects of nicotine, making it a secondary reinforcer for smoking.^{13 14} However, as the tobacco industry has long recognised, throat hit/harshness might serve as a barrier to use for nicotine-naïve youth. Describing how to design a cigarette for youth, one internal 1973 R.J. Reynolds memorandum¹⁵ noted that ‘the beginning smoker and inhaler has a low tolerance for irritation, hence the smoke has to be as bland as possible’. It recommended ‘holding pH down, probably below 6’, a value that would render practically all the nicotine in the non-volatile protonated form.

In the context of high rates of ENDS use by underage youth, and given the industry’s ability to fine tune product harshness, the question has arisen whether public health bodies should consider regulating nicotine form to make ENDS aerosols more aversive to nicotine-naïve youth, while remaining appealing to experienced combustible tobacco consumers.^{6 16} However, as with nicotine flux,^{17 18} throat hit from ENDS products is driven by several overlapping physical processes that govern transport of nicotine gas to sensory nerve sites in the throat. These processes are impacted by variables such as the ENDS liquid composition (eg, propylene glycol to vegetable glycerin ratio (PG/VG), nicotine concentration and nicotine form), the thermodynamic properties of nicotine, the particle concentration and temperature of the inhaled aerosol (and therefore the power of the ENDS device), and user puff topography. There is currently no theoretical model in the public domain that can be used to predict how these variables interact to produce a given sensory experience. Such a model could be leveraged to elucidate strategies for regulating ENDS products in a manner which supports public health aims.

While some previous investigations have examined aspects of ENDS nicotine transport in the airways, they have not attempted to link ENDS product design and operation variables to throat hit in a comprehensive model. For example, Asgari *et al*¹⁹ and Pourhashem *et al*²⁰ used detailed computational fluid dynamics (CFD) to simulate ENDS aerosol transport in the airways and found that nicotine delivery to the airway walls was dominated by gas phase absorption, not particle-phase deposition. Other investigators used in vitro measurements^{21 22} with ENDS aerosols flowing through realistic geometries of the upper airways. For example, Solingapuram *et al*²¹ highlighted the importance of nicotine form on the retention of nicotine in the airways, where higher nicotine pH levels (ie, higher freebase nicotine fractions) corresponded to more nicotine retention.

In another theoretical study, Asgharian *et al*^{23 24} examined ENDS aerosol transport in the oral cavity and found that gas absorption was the dominant mechanism of nicotine transport to the airway walls and was sensitive to the assumed nicotine activity coefficient. Activity coefficient characterises the degree of deviation of the vapour pressure of a volatile compound in a liquid solution (eg, nicotine in propylene glycol, glycerin and water) from ideal solution behaviour, that is, neglecting attractive or repulsive force interactions between molecules of different species in the solution. A nicotine activity coefficient greater than unity in an ENDS aerosol particle would result in greater gas phase concentration of nicotine surrounding

the particle than would be computed based on the properties of pure nicotine (see online supplemental equation S5). The activity coefficient of nicotine in an ENDS aerosol is a function of particle composition, and therefore changes as a particle picks up moisture while flowing through the airways.

While these studies provide insights on nicotine absorption mechanisms in the upper airways, there remains a need to characterise and simulate the processes governing nicotine absorption and throat hit over a wide range of ENDS powers, liquid compositions and user puffing topography. In this study, we sought to formulate and code a first-principles theoretical model that can be used to predict throat hit quantitatively as a function of key ENDS variables. In particular, this model couples physical processes governing ENDS device emissions to the mechanics of the upper airways to compute nicotine gas absorption in the throat. We compare model predictions to reported throat harshness from two clinical studies conducted by our group^{25 26} to examine the degree to which computed absorption in the throat predicts subjective harshness when ENDS electrical power, nicotine concentration and nicotine form are manipulated.

METHODS

Problem description and model overview

We seek to model gas phase nicotine absorption in the throat airway when a person inhales from an ENDS product with known geometry, electrical power, as well as concentrations of propylene glycol (PG), vegetable glycerin (VG), freebase nicotine and protonated nicotine. This computation requires simulating both the aerosol production process in the ENDS device and the heat and mass transfer processes that are involved as the aerosol travels first through the ENDS mouthpiece, and then the mouth and throat. The composite system modelled is illustrated in figure 1.

When the relatively warm aerosol enters the mouth, it encounters a flow boundary with an assumed temperature of 37.5°C and 100% relative humidity. The difference in temperature and humidity between the airway walls and the incoming aerosol creates thermal and mass concentration gradients that drive the transport of species and thermal energy between the airway walls and the aerosol gas phase, and, subsequently, between the aerosol gas and particle phases. The aerosol particles absorb water and cool down as they pass through the mouth and throat, while the airway walls absorb thermal energy and gas phase nicotine, PG and VG. By denuding these compounds from the gas phase, the walls induce evaporation of these compounds from the particles, modifying the composition of the particles as they flow through the airways. These processes are illustrated in online supplemental figure S1.

We simulate these processes by computing the instantaneous position, temperature and composition of the aerosol emitted by an ENDS device during a puff. The temperature and composition of the gases and particles are then tracked in time as the aerosol travels through the mouth and throat, exchanging thermal energy and species such as nicotine, water, PG and VG with the airway walls. The inhalation pattern is obtained from measurements of puff topography (see next subsection). Each puff is treated as a one-step process that takes the aerosol directly from the ENDS device into the airways, simulating what ENDS users commonly refer to as ‘direct-to-lung vaping’.

The mouth–throat geometry used in this study was obtained from a CT scan of a 50-year-old volunteer and is described in Hosseini *et al*.²² The modelled airway domain extends from the lips to the upper trachea, and the flow in it is treated as

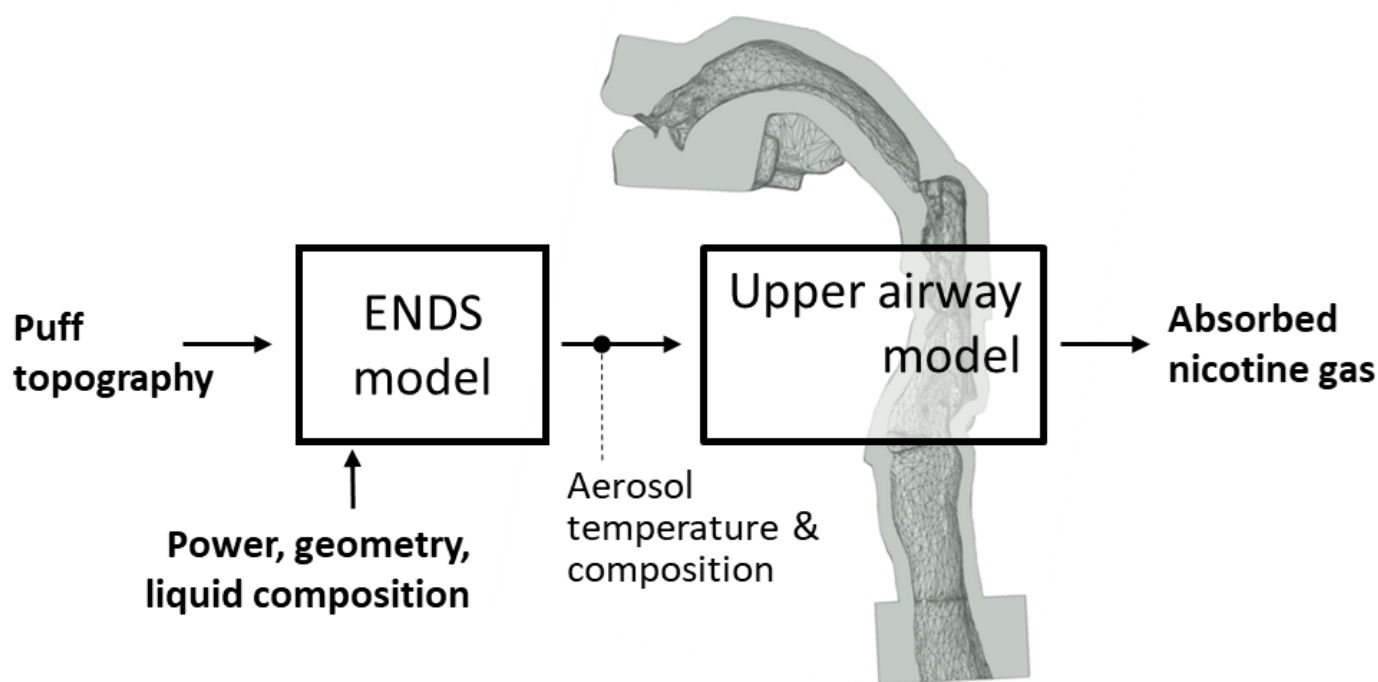


Figure 1 Schematic representation of the modelled system consisting of the electronic nicotine delivery systems (ENDS) device and upper airway.

one-dimensional. The mouth–throat is discretised into 66 segments along the flow direction (see online supplemental figures S2 and S3), and at each airway segment the heat and mass exchange between the aerosol bulk phase and airway walls is computed, and the temperature and composition of the aerosol exiting the segment updated accordingly. In this manner, the absorption of nicotine gas by the airway can be computed for each segment of the domain, for every puff drawn through it.

The ENDS emissions model used in this work is detailed in Talih *et al*,²⁷ and takes as inputs the device power, liquid composition, device geometry and user puff topography and computes the time-resolved temperature, mass concentration and composition of the multicomponent aerosol that exits the heating coil/wick assembly.

To compute the temperature of the aerosol exiting the ENDS mouthpiece, we coupled the ENDS model of Talih *et al*²⁷ to a boundary-layer heat transfer model using the geometry of the Subox Mini-C ENDS product used in the clinical studies described below (see online supplemental figure S4). The exit temperature was verified by measurements described in the online supplemental information section and shown in online supplemental figure S5.

ENDS aerosol particles are small enough to be treated as quiescent with respect to the surrounding gases, and the heat and mass transfer between the gas and particle phases is sufficiently small that it can be computed using Maxwell's quasistationary diffusion equation.²⁸ Particle deposition in the upper airways is neglected.

Energy and mass exchanges between the bulk and airway wall are computed at each segment using a boundary layer formulation in which the local heat transfer coefficient is determined as a function of flow rate by fitting to a 3D CFD simulation (see online supplemental figure S6). Mass transfer coefficients were determined from the heat transfer coefficient using the heat-mass transfer analogy.²⁹ The governing equations and solution method, as well as additional details regarding airway

discretization and CFD simulations, are given in the online supplemental information.

Human subjective effects

We retrieved data from two recent clinical studies performed by our group at Virginia Commonwealth University in which subjective 'throat hit' was reported following ENDS use (see in Hiler *et al*²⁶ and Eversole *et al*²⁵ for more details). In brief, these studies involved ENDS liquid compositions with freebase or protonated nicotine, varying nicotine concentration and varying electrical power, as shown in table 1. Both studies used a SUBOX Mini-C ENDS. In the first study (S1),²⁶ 32 participants used the device under 4 conditions differing by power and liquid nicotine concentration. For all conditions of study S1, nicotine was in the freebase form. In each condition, participants used the ENDS device for a 10-puff directed bout followed by a 60 min ad lib smoking period. In the second study (S2),²⁵ 23 ENDS users were recruited, but with the nicotine being 95% protonated.

Outcomes of both studies included puff topography (ie, number of puffs, puff duration, interpuff interval, and flow rate) and various subjective effects including throat hit. 'Throat hit' was reported on a visual analogue scale ranging from 0 (minimum) to 100 (maximum). For both studies, the 10-puff directed bout was used for correlation against model predictions of nicotine absorption in the throat. Model simulations were run for each participant and use condition using the measured puff topography records of each participant, resulting in a total of $n=127$ and $n=106$ simulations for studies S1 and S2, respectively. For each participant, we computed the total nicotine gas absorbed in the oral cavity (from the lips to the oropharynx) and the total nicotine absorbed in the throat (oropharynx to the trachea) during the 10-puff bout, using the average flow rate and puff duration of each individual user. The regional nicotine absorption was computed by summing the simulated wall nicotine absorption at each discrete flow

Table 1 Model input and output variables of studies S1 and S2

Condition	Model inputs			Model outputs				
	Device conditions	Puff topography		Nicotine flux	Nicotine inhaled		Nicotine absorbed	
	Nic Conc./power	Flow	Duration		Total	Gas	Oral cavity	Throat
	(mg/mL)/(W)	(L/min)	(s)					
S1-1 (n=32)	3/13.5	7.3 (3.2)	4.0 (1.6)	7.2 (4)	34 (31)	1.1 (0.8)	0.6 (0.3)	0.9 (0.5)
S1-2 (n=31)	8/13.5	6.9 (2.5)	3.3 (1.1)	16 (10)	64 (55)	2.2 (1.2)	1.2 (0.5)	1.8 (0.8)
S1-3 (n=32)	3/40.5	10.6 (3.6)	2.9 (1.0)	47 (9)	146 (71)	3.1 (1.9)	1.0 (0.5)	1.0 (0.5)
S1-4 (n=32)	8/40.5	9.7 (3.3)	2.3 (0.8)	114 (28)	282 (183)	5.6 (3.7)	1.9 (1.2)	2.0 (0.8)
S2-1 (n=17)	10/15	10.9 (4.4)	4.2 (1.1)	36 (12)	162 (89)	0.3 (0.1)	0.1 (0.05)	0.2 (0.1)
S2-2 (n=17)	15/15	10.8 (3.8)	3.8 (1.3)	48 (19)	206 (145)	0.4 (0.2)	0.2 (0.1)	0.3 (0.1)
S2-3 (n=17)	30/15	11.2 (3.6)	3.4 (0.9)	91 (34)	337 (217)	0.6 (0.3)	0.3 (0.1)	0.5 (0.2)
S2-4 (n=17)	10/30	10.9 (3.6)	2.8 (0.7)	95 (20)	281 (114)	0.3 (0.1)	0.1 (0.04)	0.2 (0.05)
S2-5 (n=20)	15/30	10.4 (4.1)	2.4 (0.6)	131 (33)	337 (145)	0.4 (0.2)	0.1 (0.1)	0.2 (0.1)
S2-6 (n=18)	30/30	10.4 (3.9)	2.4 (0.8)	270 (78)	713 (410)	0.7 (0.4)	0.3 (0.1)	0.4 (0.2)
Model inputs include liquid nicotine concentration, device power and puff topography. Propylene glycol to vegetable glycerin ratio was 30/70 for both studies. Nicotine was in the freebase form for study S1, and 95% in the protonated form for study S2. Model outputs included nicotine inhaled by the user, nicotine flux (nicotine inhaled per puff second) and gas phase nicotine absorption in both the oral cavity and throat regions of the upper airways. Model input results are presented as mean (SD) and model output are reported as mean (SD).								
Nic Conc., nicotine concentration; S1, first study; S2, second study.								

segment in the oral cavity or the throat at each time step of the simulation.

Influence of ENDS liquid, power and flow rate on throat nicotine absorption

To elucidate the influence of ENDS use variables separately, we ran several computer simulations in which device power, nicotine concentration, freebase nicotine fraction, PG/VG ratio and flow rate were varied one at a time relative to a baseline condition. The value of the selected variable in each condition was chosen to be within the range of the range of values in the clinical studies, except for PG/VG ratio. The latter was varied between 70/30 and 30/70, in line with commonly found values in commercially available products. In each simulation, we computed the mass of nicotine absorbed in the oral cavity and laryngeal tracheal airways and the local nicotine gas absorption flux ($\mu\text{g/s}\cdot\text{cm}^2$) throughout the mouth–throat. For each simulation, we also tracked in detail how temperature, particle water uptake, mass exchange at the airway wall boundaries and nicotine activity coefficient varied as the aerosol traversed the airways. The input conditions of these simulations are given in [table 2](#).

Statistical analysis

Linear regression of predicted and measured variables was performed using SPSS V.23.0. Statistical significance was set at $p<0.05$. Normality for regression analysis of subjective throat harshness vs predicted nicotine vapour deposition ([figure 2](#)) was assessed using the Shapiro-Wilk test, which showed slight deviation from normality around the tails. Therefore, we used a bootstrap method^{30 31} to compute the Pearson r correlation coefficient for each of 10 000 resampled datasets. The mean and 95% CI for r were reported.

RESULTS

Predicted throat nicotine gas absorption and reported throat hit

[Table 1](#) shows the predicted gas and total (particle+gas) nicotine intake and nicotine gas absorption in the oral cavity and throat across clinical study conditions. The amounts of nicotine delivered to the throat surfaces in the freebase conditions (S1) were 5–10 times greater than the amounts delivered in the protonated nicotine conditions (S2) and were unrelated to the amount of nicotine emitted by the ENDS. For example, the ENDS device in condition S2–5 had the second largest computed nicotine flux

Table 2 Predicted nicotine gas absorption in the upper airway during one 4 s puff for five hypothetical conditions that differ by power, nicotine concentration, freebase fraction, PG/VG and inhalation flow rate

Condition	Model inputs				Model outputs			
	Power (W)	Nic. (mg/mL)/FB fraction	PG/VG (v/v)	Flow rate (L/min)	Nicotine inhaled		Nicotine absorbed	
					Particle (µg)	Gas (µg)	Oral cavity (µg)	Throat (µg)
A (base case)	30	10/0.9	30/70	10	478	24.7	6.7	5.3
B	15	10/0.9	30/70	10	144	8.5	3.1	4.1
C	30	5/0.9	30/70	10	233	12.7	3.4	2.7
D	30	10/0.9	30/70	5	526	22.5	6.9	3.7
E	30	10/0.1	30/70	10	472	2.7	0.7	0.6
F	30	10/0.9	70/30	10	556	11.1	3.0	2.6

Model input bold type indicates variable manipulated relative to base case (condition A).
FB, freebase; Nic., nicotine; PG/VG, propylene glycol to vegetable glycerin ratio.

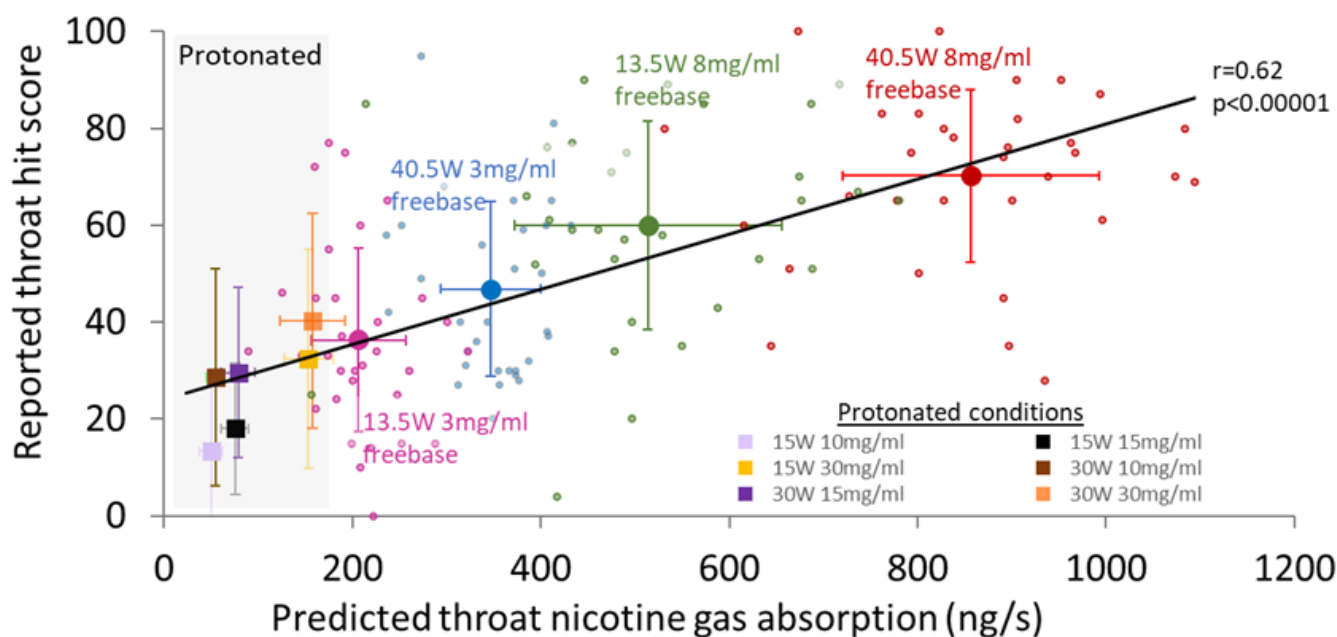


Figure 2 Subjective throat hit vs model predicted throat nicotine gas absorption across all conditions of studies of S1 (n=127) and S2 (n=106). Large, filled symbols indicate mean reported throat hit and predicted nicotine gas absorption for each condition/study; error bars represent one SD in subjective throat hit score and predicted gas absorption. Small, filled circles represent individual users in study 1; study 2 are not shown for readability. The linear regression line was fitted to the data produced by all participants and conditions in both studies.

($131 \pm 33 \mu\text{g/s}$) and yield ($337 \pm 145 \mu\text{g}$) but the smallest throat nicotine absorption ($0.2 \pm 0.1 \mu\text{g}$) of all the conditions examined. Conversely, the ENDS device in condition S1–2 yielded very little nicotine ($64 \pm 55 \mu\text{g}$) but resulted in among the highest high mean throat nicotine absorption ($1.8 \pm 0.8 \mu\text{g}$). Most importantly, computed nicotine throat absorption was significantly associated with reported subjective throat harshness ($r=0.62$, CI (0.54, 0.69), $p<0.0001$), as illustrated in figure 2, where every 100 ng/s increase in gaseous nicotine throat absorption resulted in a 5.7-point increase in throat hit score. In contrast, computed nicotine yield and flux were not associated with throat hit score ($p>0.5$).

Effects of device parameters and puffing behaviour on nicotine throat absorption

We found that throat nicotine gas absorption increased with device power, nicotine concentration, freebase fraction, VG fraction and flow rate. As shown in table 2 and illustrated in online supplemental figure S7, we found that the oral cavity absorbed more nicotine than the throat except at the lower power (condition B). Power plays an important role in the dynamics of nicotine uptake in the upper airways because it largely determines whether the particle phase—where most of the aerosolized nicotine resides—acts as a source or a sink of nicotine gas as the aerosol traverses the airways.

At lower powers, the aerosol particle mass concentration and temperature are lower (figure 3A). As a result, more water vapour is taken up by the particles from the airways and, due to the lower particle concentration, that water uptake causes a proportionally greater increase in particle phase water mole fraction (see figure 3E) per online supplemental equation S4. The greater water mole fraction in turn translates to greater nicotine

activity coefficient and greater drive for the particle phase to off-gas nicotine (see figure 3B and online supplemental figure S8). As a result, although the nicotine flux and initial gas concentration of the baseline condition (A) is much greater than that of the lower power condition (B), the nicotine gas absorption by the throat at the higher power condition is only 1.3 times greater. This result illustrates how power manipulation is one mechanism by which nicotine inhaled and throat hit can be decoupled in practical ENDS devices; a lower power ENDS can emit less nicotine but with similar throat hit/harshness as a higher power device.

Turning to the other variables, decreasing the liquid nicotine concentration (condition C) or freebase nicotine fraction (condition E) led to a proportional decrease in the predicted throat nicotine gas absorption relative to baseline. On the other hand, reducing the inhalation flow rate by half (condition D) resulted in approximately a 30% decrease in throat nicotine gas absorption relative to baseline. At the lower flow rate, the aerosol temperature is greater, causing an increase in nicotine gas concentration. While greater gas concentration increases absorption by the throat, the lower flow rate also has the effect of reducing the mass transfer coefficient in the airways. These effects tend to cancel, meaning that throat nicotine absorption is not strongly impacted by flow rate.

Finally, increasing PG content (condition F) results in lower computed nicotine absorption in the throat. One reason for the reduction is that the lower boiling temperature of PG results in lower aerosol temperatures at the exit of the ENDS mouthpiece, and therefore lower nicotine gas concentrations at the oral cavity entrance. Another reason is that the activity coefficient of nicotine decreases with increasing PG content (see figure 3B), decreasing the equilibrium mass concentration of nicotine gas at

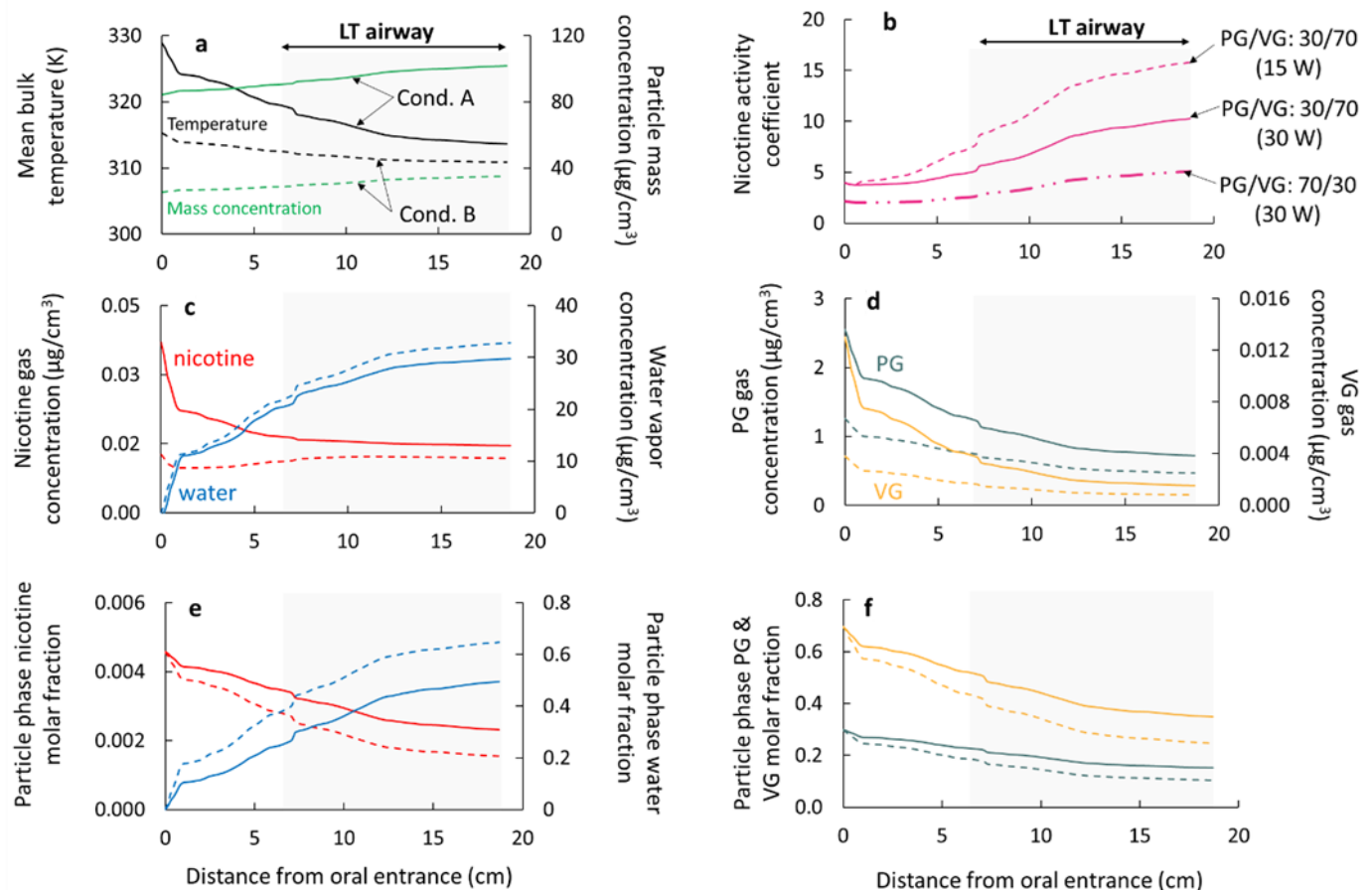


Figure 3 Bulk temperature (a), nicotine activity coefficient (b), vapour phase concentration (c, d) and particle phase molar fractions (e, f) variation along the upper airway. Solid lines correspond to condition A, dashed lines to condition B and dash-dotted lines to condition F. 'LT' stands for laryngeal–tracheal, or throat region. PG, propylene glycol; VG, vegetable glycerin.

the particle surfaces. These results are qualitatively in line with those experimentally reported by Hosseini *et al.*,²² where liquids of higher VG content resulted in higher nicotine uptake in the upper airways. We note, however, that previous clinical studies had contradictory findings on the effect of PG/VG ratio on throat sensation,^{32–35} possibly due to power and nicotine formulation differences.

Phenomenological overview and summary

Online supplemental figure S9 qualitatively maps the direction and strength of the effects of ENDS power and liquid variables on nicotine absorption in the throat, as deduced from the simulations. As shown, the mass transfer rate of gas phase nicotine to the airway walls is governed by two core parameters: the gas phase nicotine concentration and the mass transfer coefficient (see online supplemental equation S2). As the product of these two parameters increased or decreased, so did the relative rate of nicotine absorption by the upper airways.

Perhaps the most striking feature that emerged from the simulations was the degree to which throat absorption was influenced by the nicotine activity coefficient. Activity coefficient varied by an order of magnitude across conditions and within condition as the aerosol traversed the airways due to its strong coupling to particle phase water content and, secondarily, to particle phase PG/VG ratio. This variation is responsible for the outcome that increasing ENDS power results in lower throat nicotine

absorption per unit nicotine emitted by the ENDS, and likely explains why participants reported the throat hit of the low power condition of S1–2 (13.5 W) to be similar to the throat hit in the high power condition S1–4 (40 W), though the latter had four times the nicotine emitted and seven times the nicotine flux.

Apart from the previously discussed effects of liquid nicotine concentration and freebase nicotine fraction on gas phase concentration, another impactful variable was found to be the PG/VG ratio of the liquid. The dominant effect of increasing PG/VG ratio was that it reduced the nicotine activity coefficient, and therefore the nicotine gas concentration (eg, a dry 100/0 PG/VG aerosol exhibits approximately 1/5th the gas phase nicotine concentration of a dry 0/100 PG/VG aerosol, all else being equal; see online supplemental table S1). Increasing PG/VG ratio also decreased the aerosol temperature and increased the aerosol concentration; both of these secondary effects reduced gas phase nicotine concentration and throat absorption. Finally, increasing flow rate reduced aerosol temperature at the exit of the ENDS mouthpiece, resulting in a reduction in nicotine gas concentration. The relative effect of reducing nicotine gas concentration versus increasing the mass transfer coefficient depended on the particular conditions simulated. Among other factors, the mass transfer coefficient is a strong function of fluid motion near the flow boundaries; as the flow velocity increases the fluid near the flow boundary (ie, the airway wall) undergoes more shear and increases the concentration gradients that drive heat and mass transfer.

DISCUSSION AND CONCLUSIONS

For decades, the tobacco industry has engineered the sensory characteristics of tobacco products, including throat hit. People who use tobacco products associate throat hit with the positive psychological effects of nicotine, making it a secondary reinforcer for smoking. On the other hand, high throat hit could make products aversive to naïve nicotine users and serve as a barrier to consumption for this population. While the roll-out of protonated nicotine products by the ENDS industry in 2015 is widely regarded as a key factor for rapid youth uptake in subsequent years (eg, Gholap *et al*³⁶ and Glantz *et al*¹), other factors influencing throat harshness, including interactions between ENDS liquid composition, puffing behaviour and device power have not been systematically mapped in the open literature. This study presents a comprehensive model, grounded in first principles, that describes and connects the various physical processes that govern ENDS operation, emissions and nicotine vapour deposition in the throat. While a limitation of this study is that the mouth–throat geometry used in the simulations was derived from a CT scan of the airways of one 50-year-old individual, we found that over a broad range of ENDS conditions involving 55 individuals, predicted nicotine throat absorption was strongly correlated to subjective throat harshness reported by those individuals.

In addition to nicotine form, we found that nicotine absorption in the upper airways is influenced by interactions among PG/VG ratio, nicotine concentration and device power. One striking finding was that though reducing device power proportionally reduces nicotine emitted by the ENDS, it has far less influence on nicotine absorption in the throat, and presumably, on throat harshness. In addition to nicotine form, power is therefore a lever by which nicotine flux can be decoupled from throat hit, one that should be kept in sight if product harshness is regulated in the future. The simulations also highlighted that nicotine activity strongly influences absorption in the airways, and that activity changes drastically due to water uptake and other compositional changes as the aerosol traverses the airways.

In conclusion, this study demonstrates that subjective nicotine throat hit/harshness ratings from participants using an ENDS are associated reliably with ENDS device and liquid properties using a simplified physics-based mathematical model of aerosol generation and transport. We found that the computed rate of nicotine gas uptake by the airway walls during puffing drives throat harshness and that this process is governed primarily by three independently controlled variables: ENDS device power, liquid nicotine concentration and nicotine protonation. This study suggests the basic technical feasibility of providing targets for both nicotine flux and throat hit in ENDS product regulation, for example, with the aim of providing cigarette smokers a satisfying alternative whose harshness non-smokers find aversive.

Contributors Conceptualisation: AS and MEH. Mathematical model development: MEH, AS and ST. Computational fluid dynamics simulations: MA and IL. Activity coefficient simulations: CA. Upper airway geometry: LG and SH. Manuscript drafting: MEH and AS. Manuscript revision: MEH, AS, RS, LG, TE, AB, DA and IL. AS is the guarantor.

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Competing interests TE and AS have been paid consultants in litigation against the tobacco industry and also the electronic cigarette industry and are named on one patent for a device that measures the puffing behaviour of electronic cigarette users

named on one patent for a device that measures the puffing behaviour of electronic cigarette users, on another patent application for a smartphone app that determines electronic cigarette device and liquid characteristics, and a third patent application for a smoking cessation intervention. TE is also named on a patent for a smartphone app that determines electronic cigarette device and liquid characteristics.

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Ethics approval Not applicable.

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